

in both the user device and the satellite, keeping the beam aligned when the user moves, additional optical circuitry, optics/electronics interfaces and so on. These are not just challenges but involve significantly more cost and complexities especially in user devices. Also, key proposals of sharing the spectrum with mobile systems for access will not be possible," explains Dr Borkar.

High-altitude drones. Facebook's drone idea sounds very similar to Google's balloons. Just that they chose drones instead of balloons because they have more endurance and can also be steered more accurately. At an altitude of around 20km above ground, solar-powered drones would be able to cruise easily and conserve power, and broadcast signals powerful enough to cover a city-size territory with a medium population density.

A high-profile team is working to finalise the communication equipment payload of the drones and also figure out how FSOs can be used to make the solution more effective. Once again, Dr Borkar helps tone down our expectations: "The drones will be flying at lower levels than the satellites. Hence, the number of drones will be significantly larger than satellites to provide continuous coverage, which will likely be approximately six or seven times. This creates major issues including path control for drones, handovers and network management."

Peak or trough, satellites will reach you

Since decades, satellites have proved their usefulness in connecting remote areas. But, due to the heavy costs involved, these have been used more for military and commercial purposes than for civilian communications. Plus, traditionally, geostationary-Earth orbit (GEO) based systems have had major latency issues because of the large distance of links, making these

India is the third-largest Internet user base in the world, comprising around eight per cent of the world's Internet users, which is less than 20 per cent of our population (Courtesy: www.internetlivestats.com/internet-users-by-country)

useless for real-time applications like voice-conferencing, video-conferencing, machine-to-machine (M2M) communications and cloud computing.

However, in recent times, companies like SpaceX, OneWeb and O3b Networks have started promising affordable, high-speed communications with satellites in low- and medium-Earth orbits (LEO and MEO, respectively).

SpaceX. This company hopes to put in place a 4000-satellite network in LEO by 2030, and if this really happens, all you would need is a handy receiver to connect from just about any spot on the Earth.

Satellites have always been costly business, and SpaceX plans to change this. One of the major cost-points is the rocket, which goes up in flames on re-entry, so it can be used only once. So the company's scientists are now trying to develop a reusable rocket, which puts the satellite in space and returns to a barge, from where it can be safely steered to shore and reused. They also plan to make smaller, lighter satellites, so that the overall cost will make more economic sense than laying fibre-optic cables.

OneWeb. OneWeb, backed by big names like Virgin Group, Qualcomm, Airbus Group, Coca Cola and Bharti Enterprises, has its own share of game-changing plans to launch



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